

LAB NOTES · ISSUE 04 OF 04

Lab: synthetic datasets for MLOps and GenAI practice.

Synthetic data, real questions. No customer data anywhere.

04

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Swipe →

Model monitoring dataset

01 What it simulates

A churn model's predictions across regions and channels.

02 Columns

Model version, score, actual label, latency, errors.

03 Drift periods

Before and after windows to detect degradation.



RAG evaluation dataset

01 What it simulates

RAG answers across prompts and domains.

02 Columns

Prompt template, retrieval top-k, groundedness, latency.

03 Labels

Pass / fail and hallucination risk.

Questions to practise

01 Monitoring

Did accuracy degrade after the drift period?

02 Versions

Which model version performs best, and where?

03 RAG

Which prompt template has the best pass rate?

04 Trade-off

How does top-k affect latency and groundedness?

How I use them

- 1 Generate with a fixed seed
- 2 Publish on Kaggle
- 3 Analyse with pandas or BigQuery
- 4 Build a dashboard
- 5 Walk through it in interviews

The generator script ships with the data, so anyone can reproduce it.

HONEST LABEL

| Synthetic by design.

CC0, seeded, no real customer, system or billing data.



KEY TAKEAWAY

Practise monitoring and evaluation on data you can share.

- Seeded and reproducible
- Monitoring drift and RAG quality
- Real interview questions

YOUR TURN

What dataset do you wish existed for MLOps practice?

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